

Incomplete reversibility of fixation-induced changes in postmortem human brain WM MRI following rehydration



Lennart Bedarf^{1,2}, Dominique Neuhaus^{1,2},
Eva Scheurer^{1,2}, Claudia Lenz^{1,2}

¹Institute of Forensic Medicine, Department of Biomedical Engineering, University of Basel, Switzerland
²Institute of Forensic Medicine, Health Department Basel-Stadt, Basel, Switzerland



Background and Objective

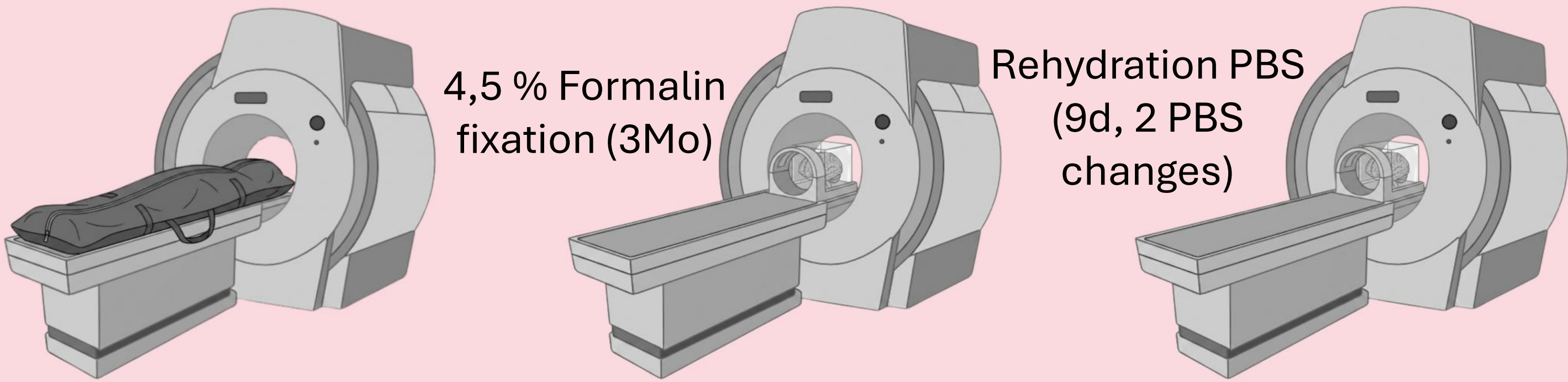
- Validation of MRI biomarkers: based on formalin-fixed ex situ imaging and histology
- Fixation: alters relaxation properties and tissue anisotropy¹⁻⁴
- Rehydration: suspected to partially reverse fixation effects → irreversible protein cross-linking suggests incomplete reversibility⁴⁻⁵

Objective: Assess effects of formalin fixation and rehydration on MRI derived measures in human brain WM

Methods

Subjects & MRI:

- Two human brains scanned on Siemens Prisma 3T



MRI Protocol:

- MP2RAGE: structural imaging and tissue segmentation
- Multi-echo GRE (12 echoes): R_2^* mapping
- Multi-shell DTI ($b = 0, 1000, 2000, 3000 \text{ s/mm}^2$)

Image processing:

- HD-BET⁶ - brain extraction, FSL FAST⁷⁻¹⁰ - segmentation, FSL eddy_correct⁷⁻⁹, FSL DTIFIT⁷⁻⁹ - MD, FA, AD, RD

Analysis:

- Fiber angles derived from DTI V1, binned $18 \times 5^\circ$ ($0^\circ - 90^\circ$)
- R_2^* calculated voxel-wise, data normalised and registered
- R_2^* anisotropy index $AI = \frac{(R_2^{*max} - R_2^{*min})}{(R_2^{*max} + R_2^{*min})}$

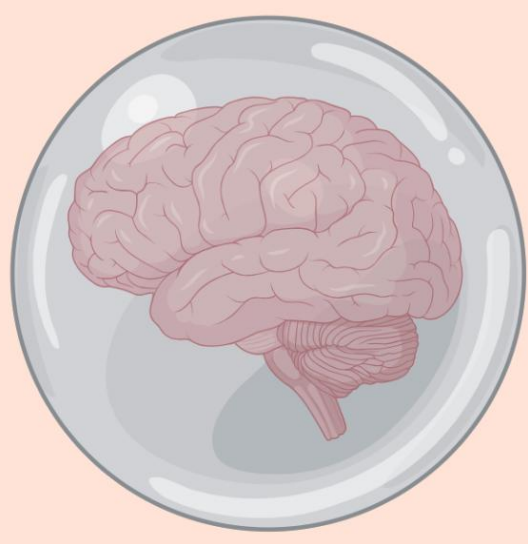
Conclusion

Fixation and rehydration:

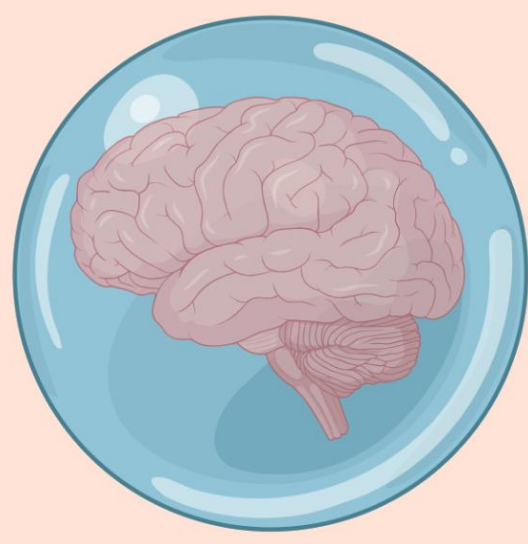
- substantially alter MRI-derived measures
- heterogeneous & partial reversibility



In situ



Formalin fixed



Rehydrated

Comparison of in vivo MRI to ex situ measurements for biomarker validation is limited

Results and Discussion

- Formalin fixation alters WM microstructure
- Fixation: FA ↓, MD ↓, AD ↓, RD ↓, AI ↓, $R_2^* \uparrow$
- Rehydration: FA ↓, MD ↑, AD ↑, RD ↑, AI ↑, $R_2^* \downarrow$
- Angular dependency: in situ > rehydrated > fixed

Table 1: Diffusion parameters FA, MD, AD and RD, as well as R_2^* and anisotropy index across the three tissue conditions.

	In situ	PM ex situ formalin-fixed	PM ex situ rehydrated
FA	0.47 ± 0.02	0.37 ± 0.05	0.30 ± 0.00
MD [$10^{-5} \text{ mm}^2/\text{s}$]	16.22 ± 6.48	13.03 ± 1.34	15.26 ± 2.84
AD [$10^{-5} \text{ mm}^2/\text{s}$]	24.80 ± 9.82	17.54 ± 2.23	19.87 ± 4.01
RD [$10^{-5} \text{ mm}^2/\text{s}$]	12.10 ± 4.71	11.01 ± 1.09	12.95 ± 2.25
$R_2^* [\text{s}^{-1}]$	27.79 ± 4.48	32.39 ± 0.38	28.40 ± 0.40
AI [%]	9.3 ± 3.3	1.5 ± 0.0	2.8 ± 0.4

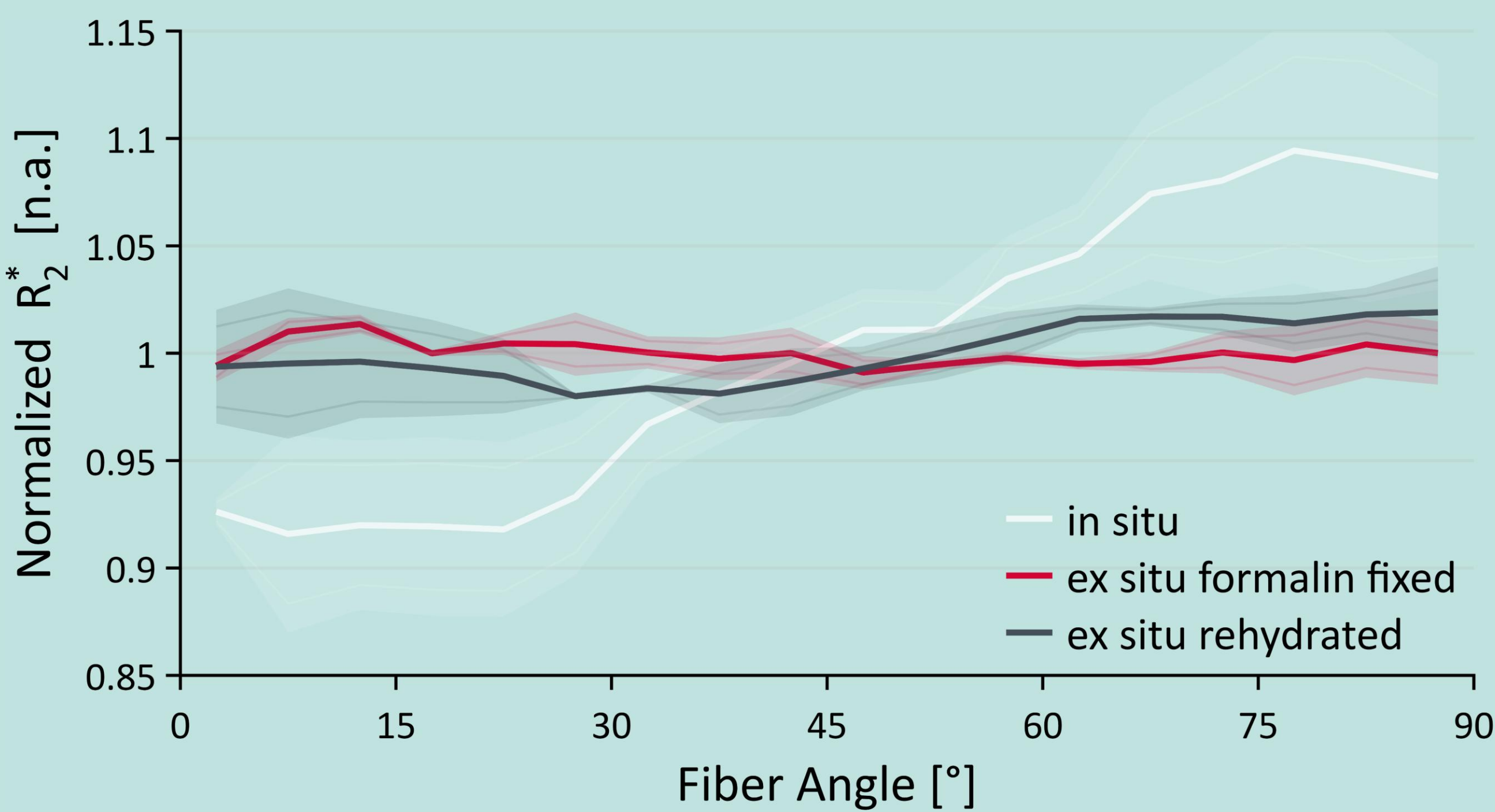


Figure 1: Normalized R_2^* values across the three tissue conditions over the calculated tissue fiber angle.

Formalin fixation:

- Diffusivity, FA and R_2^* orientation dependency: ↓

Rehydration:

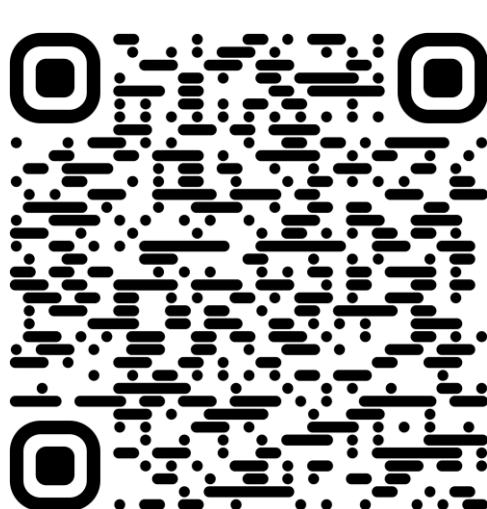
- partial recovery of MD, AD, AU and R_2^*
- FA remained reduced, RD exceeded in situ → Tissue properties not fully restored

Key Message: Fixation effects are partly reversible, but not fully recoverable through rehydration

- Osmotic shifts & water distribution may cause apparent («pseudo-») recovery
- Fixation-induced changes should be considered when validating in vivo biomarkers using ex situ MRI and histology

[1] Shatil et al. *Front Med*. 2018
[2] Birkel et al. *Magn Reson Med*. 2018
[3] Birkel et al. *NMR Biomed*. 2016
[4] Neuhaus et al. *Magn Reson Med*. 2024
[5] Schmieder et al. *Magn Reson Med*. 2018
[6] Isensee et al. *Hum Brain Mapp*. 2019
[7] Woolrich et al. *NeuroImage*. 2009
[8] Smith et al. *NeuroImage*. 2004
[9] Jenkinson et al. *NeuroImage*. 2012
[10] Zhang et al. *IEEE Trans Med Imaging*. 2001

Download poster



My contact information

